

**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**

Paper Code : BS-B401/BS-BIO401/BSC 401/BSC-401/BSC401 Biology

UPID : 004408

Time Allotted : 3 Hours

Full Marks :70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**1. Answer *any ten* of the following :

[1 x 10 = 10]

- (I) Discontinuous synthesis of DNA occurs in _____ during replication of DNA.
- (II) _____ amino acids can be produced by the body even if we do not get it from the food we eat.
- (III) Krebs cycle occurs in which part of the cell?
- (IV) Which microorganisms are used to produce alcohol?
- (V) The phenotypic ratio of dihybrid experiment is _____.
- (VI) _____ is also known as "suicidal bag".
- (VII) The genotypes of blood group O is _____.
- (VIII) The general molecular formula of Carbohydrates is _____.
- (IX) In _____ inhibition, the inhibitor competes with the substrate for binding to the active site of an enzyme.
- (X) The excision of introns and the formation of final mRNA molecule by joining the exons is called _____
- (XI) The chromosomes with satellite are known as _____.
- (XII) Glycolysis is also known as _____

Group-B (Short Answer Type Question)Answer *any three* of the following :

[5 x 3 = 15]

2. What are different types of culture media based on the physical state? Distinguish between agar and broth? [5]
3. Draw a comparison between eye and camera. [5]
4. Draw labeled diagram of Animal cell as seen in Electron microscope. Comment on characteristics of Animal cell. [5]
5. Write short note on Central dogma. [5]
6. Explain the types of lipids depending on the esterification. [5]

Group-C (Long Answer Type Question)Answer *any three* of the following :

[15 x 3 = 45]

7. (a) Explains the phases of microbial growth kinetics with graph. [10]
(b) Differentiate between sterilization and pasteurization. [5]
8. Explain why engineers need to study biology? [15]
9. (a) What is DNA replication? [2]
(b) State in brief the mechanism of DNA replication with diagram. [13]
10. (a) Write a short note on secondary structure of protein. [5]
(b) Define essential, conditionally essential and non-essential amino acids giving one example from each. [4]
(c) Define Monosaccharide, Disaccharide and Trisaccharide with example. Distinguish between cis fat and trans fat [6]
11. (a) What is cell cycle? [2]
(b) Describe the different phases of cell cycle. [10]
(c) What is Crossing Over? [3]

*** END OF PAPER ***